



ADNOC GAS DEFINE STAGE PROJECT

CONTRACT No. 4700024157
PROJECT No. 5.0 (P5)

MR FOR GAS POWER GENERATOR

DOCUMENT NO. ADN-AEC-ME-SPC-026

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0	10/10/2024	HS	WJ	AA	ISSUED FOR QUOTATION
REV	DATE	PREPARED	REVIEWED	APPROVED	DESCRIPTION

THIS DOCUMENT IS INTENDED FOR USE BY ADNOC ONSHORE AND ITS NOMINATED CONSULTANTS, CONTRACTORS, MANUFACTURERES AND SUPPLIERS



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ADNOC CONTRACT NO. 4700024157

PROJECT NO.
AP-044

CLASS 1

DATE: 10-10-2024

PROJECT:	AP-044	ADNOC GAS DEFINE STAGE PROJECT 5.0 (P5)			
ADNOC CONTRACT NO:	4700022685				 
DOC. No:	ADN-AEC-ME-SPC-026				
REV NO.	0				
MR FOR GAS POWER GENERATOR					
(A) Preamble					
SL. No.	Description				
1.0	INTRODUCTION:				
	ADNOC is aiming to increase oil production to produce 5.0 MMBOPD by year-end 2027. Accordingly, associated gas production is anticipated to increase. Hence, the overall gas supply scenarios for major ADNOC Gas (hereafter referred to as COMPANY) assets i.e., Das Facilities, Bab and Habshan Complex (Habshan 0, 1, 2, 4, and 5), Asab and Bu Hasa, is anticipated to change depending on their feed sources, profiles including rates, compositions, and operating conditions. COMPANY’s permanent facilities to accommodate increased gas supply by 2027 will be achieved through long term EPC Contract. As the permanent facilities will only be ready by year 2027, COMPANY intends to exercise the short-term solutions (Rental Compressor and corresponding Balance of Plant (BOP)) for Buhasa plant to accommodate peak gas flow and ensure bottlenecks in years 2025, 2026 and 2027 are mitigated, prior to implementation of a long-term solution that caters the entire production profile.				
2.0	PROJECT OBJECTIVE				
	The short-term solution includes Rental Compressor (By ‘Design Built Own Operate and Maintain’ - DBOOM Contractor) and corresponding Balance of Plant (BOP) works for impacted COMPANY facilities is as follows. DBOOM Scope – Design, Build, Own, Operate and Maintain (DBOOM) the Rental Compression facility including required utility to process additional gas from Buhasa P5 facility. The scope includes detailed engineering, procurement & supply, installation & testing, Commissioning and Operation & Maintenance works. BOP Scope – The Balance of Plant (BOP) facility to interface and hook up the rental compressors to the main plant production stream including New LP and IP Pipelines with associated engineering. The COMPANY shall execute the separate contract for this Scope.				
3.0	PURPOSE OF THIS DOCUMENT				
	The purpose of this document is to define the minimum requirements for the Design, Engineering, Material Procurement Manufacture / Fabrication, Assembly, Testing, Inspection, Performance Testing, Painting, Documentation, Transportation Packing, Forwarding and Commissioning of GAS ENGINE DRIVEN GENERATORS for the P5 Project. This document along with the referenced documents and standards are binding and are to be complied by the VENDOR. Also VENDOR shall make an unequivocal statement that the GAS ENGINE DRIVEN GENERATORS offered are approved for Installation in Abu Dhabi, UAE.				
4.0	SYSTEM DESCRIPTION				
	P5 Compression Facility shall be fed from the natural gas generator [GEG] G-4101-01A & G-4101-01B and one number emergency diesel generator. a. During normal operation, one of the GEG shall run and feed power to the facility. b. If a failure occurs during a normal operation, the EDG shall be switched on. c. GEG-2 shall be switched and synchronized with edg and once synchronized. d. Any generator running parallel shall be synchronized prior to the parallel operation, this shall be made sure by the generator vendor with his control system.				
5.0	VENDOR SCOPE OF SUPPLY				
	Vendor shall supply Gas Engine Driven Generator set with Control system, Cooling system, Connection to natural gas system, Generator set accessories, Mounting system, Base slab and vibration isolation, Controller Panels alongwith synchronization relays, Weatherproof housing and all other accessories to complete the power generation requirement mentioned in the detailed requirements below. 0.525 MW, 415V, 3Phase, 50Hz, Containerized type Gas Engine Generator - 2 Nos.				
6.0	LIST OF APPLICABLE INTERNATIONAL STANDARDS				
	STANDARD REFERENCE		Compliance	Vendor Response	Remarks
	ANSI AWS D1.1/D1.1M	Structural Welding Code - Steel	Required		
	BS 4999-140	General requirements for rotating electrical machines Part 140. Voltage regulation and parallel operation of A.C. synchronous generators	Required		
	IEC 60034 (part 1, 5, 6, 8)	Rotating Electrical Machines	Required		
	IEC 60071-1	Insulation co-ordination – Part 1: Definitions, principles and rules	Required		
	IEC 60255	Measuring relays and protection equipment	Required		
	IEC 60445	Basic and safety principles for man-machine interface, marking and identification – Identification of equipment terminals, conductor terminations and conductors	Required		
	IEC 60529	Degrees of protection provided by enclosures (IP code)	Required		
	IEC 60898	Electrical accessories - Circuit-breakers for overcurrent protection for household and similar installations	Required		

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	IEC 61850	Communication networks and systems in substations	Required		
	IEC 61869 (part 1, 2, 3)	Instrument transformers	Required		
	ISO 1813	Belt drives. V-ribbed belts, joined V-belts and V- belts including wide section belts and hexagonal belts. Electrical conductivity of antistatic belts: Characteristics and methods of test	Required		
	ISO 3046-(part 1, 4)	Reciprocating Internal Combustion Engines	Required		
	ISO 8528	Reciprocating Internal Combustion Engine Driven Alternating Current Generating Sets	Required		
	ISO 10816-(part 1, 5)	Mechanical vibration – Evaluation of machine vibration by measurements on nonrotating parts	Required		
	ISO 10816-6	Part 6: Reciprocating machines with power ratings above 100 kW	Required		
	ASCE 7-05	Minimum Design Loads for Building and Other Structures	Required		
	AISC 325	Steel Construction Manual	Required		
	AISC 341	Seismic Provisioning for Steel Buildings	Required		
	Refer "Attachment-4 Applicable Codes and Standards" for the full list		Required		
7.0	LIST OF ATTACHMENTS				
	Attachment-1A Fuel Gas_Sus_Summer		Required		
	Attachment-1B Fuel Gas_Sus_Winter		Required		
	Attachment-2 Emission Compliance Data		Required		
	Attachment-3 Synchronization Diagram		Required		
	Attachment-4 Applicable Codes and Standards		Required		
	Attachment-5 Approved Vendor List 21.05.2024		Required		
	Attachment-6 Technical Deviation		Required		
	Attachment-7 Load List		Required		
	Attachment-8 Typical Gas Generators IO List		Required		
	Attachment-9 VDRL		Required		

MR FOR GAS POWER GENERATOR							
SL. NO.	DESCRIPTION	UNIT	PROJECT REQUIREMENTS	VENDOR COMPLIANCE (CONFIRMED / NOT CONFIRMED)	CONTRACTOR REMARKS	CLIENT REMARKS	REMARKS
1.0	GENERAL						
1.1	Generator Rating	-	525KW, 415V, 3Phase, 4Wire, 50Hz, Containerized type Gas Engine Generator - 2 Nos. Star Connection - Duty S1	Comply			
1.2	Temperature	5-58 deg C	Vendor to confirm compliance	Comply			
1.3	Design Life	-	05 years; Bidder to comply	Comply			
1.4	Vendor Deviation Form as per Attachment-6	-	Signed / Stamped copy is required with Offer Any deviations from the PROJECT requirements shall be listed in Attachment-6. Deviations listed elsewhere will not be considered for evaluation	Noted			
1.5	Tag No's		G-4101-01A & G-4101-01B	Comply			
1.6	Weight	kg	Vendor to Furnish	28000 kg Approx			
1.7	Compliance to IEC 60034	-	Required	Comply			
1.8	Vendor & Sub-Vendors (Vendor of all items in scope) to be as per COMPANY Approved Vendor List	As per the Attachment-5 Approved Vendor List	Vendor to Confirm.	Comply			
1.9	Hazardous Area Classification	-	Safe Area	Noted			
1.10	Anchor Bolt	-	Required	Comply			
1.11	Manufacturer	-	Vendor to Furnish	ADPOWER FZCO			
1.12	Model No.	-	Vendor to Furnish	GASGEN-800			
1.13	Country of Origin	-	Vendor to Furnish	UAE			
1.14	Foundation Load Data	-	Vendor to Furnish	TBA			
1.15	Installation/fixing detail with Foundation		Vendor to Furnish	Comply			
1.16	Documents Required with Offer		Filled and Compiled This MR Sheet Detailed Vendor Standard Datasheet of all components Vendor & Sub-Vendor List with port of origin Heat & Material Balance Diagram PID Equipment List Power Auxiliary List Consumption List Equipment Vendor Standard Catalogs MTC and Certificates Once technical closure is recieved signed stamped this MR Sheet	Comply			
2.0	Technical Requirements & Scope of Supply						
2.1	Alternator	-	Vendor to Furnish	MECC ALTE - ECO43-1M/4			
2.2	Acoustic Enclosure with complete package skid	-	Required with cooling fans Enclosure shall be strong enough to withstand the CO2 release and corresponding CO2 thermal expansion	Comply			
2.3	Starting System	-	Vendor to Furnish	Electrical Start 24V DC			
2.4	Lube Oil System	-	Vendor to Furnish	Comply			
2.5	Generator Duty Type	-	S1 according to IEC 60034-1	Comply			
2.6	Generator Insulation Temperature	-	Class F and to Class B temperature rise	Comply			
2.7	Generator Winding Insulation	-	Class H	Comply			
2.8	Installation requirement	-	Outdoor Installation in Containerized	Comply			
2.9	Fire & Gas System	-	F&G Detector Requirement 1. Flammable gas detection on the combustion and ventilation air inlet- Minimum 3 quantity 2. Gas detection on the ventilation air inlet -Minimum 3 quantity 3. Flame Detector at Enclosure including Load Compartment 4. Rise of Heat Detector at Enclosure including Load Compartment 5. Fire and Gas Beacon alarm & Horn - outside Enclosure Detector design The F&G devices shall be of minimum SIL 2 certified	Comply			
2.10	Fire Protection System	-	Required (automatic fire protection system using CO2 as the extinguishing medium	Comply			
2.11	Enclosure Temperature monitoring system	-	Required	Comply			
2.12	Sustained Overload	-	Generator shall be capable of carrying a stator current of 110% of the rated current for a period of at least one hour at rated voltage, frequency and power factor.	Comply			
2.13	AVR	-	Required (microprocessor based)	Comply			
2.14	Voltage Variation	%	±10%	Comply			
2.15	Frequency Variation	%	±5%	Comply			
2.16	Power Factor	-	0.8 lagging	Comply			
2.17	I ² t	-	Vendor to Furnish	Heat Dissipation 42753 W			
2.18	Electromagnetic Compatibility	-	Accordance to IEC 61000	Comply			
2.19	Starting Batteries	-	Required	Comply			
2.20	Type of Batteries	-	As per Manufacturer standard	Sealed Lead Acid			
2.21	Battery Charger	-	Required	Comply			
2.22	Rating Plate	-	Required	Comply			
2.23	Efficiency (50 %, 75 %, and 100 % rated output)	-	Vendor to Furnish	94,5 95,6 95,3			
2.24	Generator Cooling	-	Air Cooling or Liquid Cooling (Air cooling is preferred) Vendor shall include the number of fans and wattage	Air to Liquid Cooling			
2.25	Generator Housing	-	IP 55 with TEFC enclosure (containerized)	Comply			
2.26	Neutral Earthing	-	Solidly Earthed for LV System	Comply			
2.27	Direction of rotation	-	IEC 60034-8	Comply			
2.28	RTD	-	PT100 RTD, 2 per phase (RTDs shall be connected to the DCS)	Comply			
2.29	Terminal Boxes	-	Separate terminal boxes shall be provided for: 1. Stator connections and Stator star point connections 2. Heater connections 3. Exciter connections 4. RTD connections 5. CT connection 6. Other connections	Comply			
2.30	Frequency stability	-	Governing class A2	Comply			
2.31	Gas engine	-	Vendor to Furnish	Baudouin Engine			
2.32	Air Intake System with filter and DPI and Wild-run shutdown valve	-	Vendor to Furnish (shall be part of the skid package)	Comply			
2.33	Engine Exhaust System with Silencer	-	Vendor to Furnish	Comply			
2.34	Terminal Boxes	-	Vendor to Furnish	Comply			
2.35	Engine Cooling System	-	Vendor to Furnish	Comply			
2.36	Shaft Couplings	-	Required	Flexible Coupling			
2.37	Torsional Analysis of Engine	-	Required	Comply			
2.38	Excitation system	-	Brushless type, consisting of a main AC Exciter and a permanent magnet generator direct coupled and air cooled	Comply			
2.39	Excitation - Rated Current	-	At least 110% of the excitation current at the rated output of the generator under the most stringent operating conditions	Comply			
2.40	Excitation - Rated Voltage	-	At least 110% of the rotor voltage at the rated output of the generator under the most stringent conditions	Comply			
2.41	Replacement interval of the lubricating oil	-	Vendor to Furnish	TBA			
2.42	Noise Limit	dBA	85dB(A) at 1 meter distance	Comply			

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2.43	Auxiliary Load Requirement	-	Vendor to Furnish	10kW max			
2.44	Anti-condensation Heaters	-	Required	Comply			
2.45	Engine and Generator Painting	-	As per Manufacturer standard	Comply			
2.46	Generators Synchronization	-	Generator synchronization relays shall be provided by Generator vendor as per attached Synchronization Scheme	Comply			
2.47	Gas Pressure Regulating Skid	-	Generator synchronization relays shall be provided by Generator vendor as per attached Synchronization Scheme	Comply			
2.48	The generator protection system and trip logic shall be in line with IEEE guidelines IEEE C37.101 and IEEE C37.102.	-	Vendor to confirm compliance	Comply			
2.49	Internal and External Earthing	-	Container shall be provided with internal earthing with suitably designed earth bars alongwith earthing cables, point for generator star earthing, external enclosure earthing ensuring safe and reliable operation	Comply			
2.50	Internal Lighting	-	Container shall be provided with internal lighting and external bulk head lighting on the doors	Comply			
2.51	Selective Catalytic Reduction system (SCR)	Optional Price	Vendor shall include the price based on the overall design and compliance to the project emission requirement	Comply			
2.52	Gas Genset Auxiliary Load	Refer to the attached Load List. Vendor shall review the load list and confirm the starting of the largest motor at star-delta configuration	Vendor to confirm compliance	Comply			
2.5	Gas Composition Data	Attachment-1A Fuel Gas_Sus_Summer Attachment-1B Fuel Gas_Sus_Winter	Vendor to confirm compliance	Comply			
2.5.1	H2S Content	700 ppm	Vendor to confirm compliance	Comply			
2.6	Inlet Gas Supply			Comply			
2.6.1	Pressure	Attachment-1A Fuel Gas_Sus_Summer Attachment-1B Fuel Gas_Sus_Winter	Vendor to confirm compliance	Comply			
2.6.2	Temperature	Attachment-1A Fuel Gas_Sus_Summer Attachment-1B Fuel Gas_Sus_Winter	Vendor to confirm compliance	Comply			
2.7	Ambient requirements						
2.7.1	For Gas Engine generator set	4-58 deg C	Vendor to confirm compliance				
2.8	Gas type						
2.8.1	Biogas or Natural Gas	Dry Gas	Vendor to confirm compliance	Comply			
2.8.2	Emission Data	Compliance with Emission limits as per local regulation. Refer Attachment-2 Emission Compliance Data	Vendor to confirm compliance	Comply			
2.8.3	Emission Control Equipment	As required	Vendor to confirm as per the Emission Data requirement	Comply			
2.9	Applicable Codes and Standards	Attachment-4 Applicable Codes and Standards	Vendor to confirm compliance	Comply			
3.0	Control and Protection Requirement						
3.1	Local Engine Control Panel (LCP) Placed inside the Gas Generator Container	-	Required, 1. Protection, control and metering equipment 2. AVR 3. Complete AC sub-distribution for the generator space heater, panel space heater and any other generator auxiliary items 4. Raise/lower voltage adjuster (behind the panel door) 5. Generator status, alarm and trip functions 6. Automatic starting / synchronizing facilities with manual override, synchroscope and synchronizing lamps 7. Control of switchgear remote generator CB 8. Control gear for generator set auxiliaries 9. Control relays (under-voltage, etc...)) 10. Battery backed up DC power supply distribution for the generator protection systems (may be combined with the engine protection systems) 11. Generator testing facility 12. Winding/bearing temperature monitoring facility Vendor shall provide Redundant serial communication (RS485) for interface from Gas Generator Vendor provided LCP to Control system (UCP). 13. Fire Alarm & Detection and Fire Fighting System Equipment, Panel and Interface. 14. Gas Generator Enclosure external fans power and control panel supply with builtin contactors and external cooler temperature interface.	Comply			
3.2	Remote Generator Control Panel (GCP) Placed inside the Substation	-	Required (including control and monitoring equipment and engine measurements/controls) for all the components inside the Gas Generator Enclosure alongwith external fans through the HMI installed on the panel. Vendor shall furnish details, the panel shall be installed inside the substation. Vendor shall provide Redundant serial communication (RS485) for interface from Gas Generator Vendor provided LCP to Control system (UCP).	Comply			
3.3	Protective Devices	-	All protective relays shall be of microprocessor type. The following electrical protective relays shall be provided as minimum to protect the generator. a. Reverse power (32) b. Stator winding over temperature (49) c. Time delay zero sequence over current (51N) d. Time delay voltage restrained over current (51V) e. Synchro check relay (25) f. Undervoltage relay (46) g. Negative phase sequence relay (46) h. Lockout relay (86) i. Over voltage protection (59) j. Field failure protection (loss of excitation) (40) k. Diode failure protection l. Trip circuit supervision relay (96) m. Instantaneous over current relay (50) n. Rotor earth fault (64R) o. Stator earth fault (64S) p. Frequency relay (81) q. Restricted Earth Fault Protection (87N) r. Differential Protection for Generator (87G)	Comply			

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3.4	Interface with DCS & ESD		Gas Engine Driven Generators shall be interfaced to DCS and ESD system for monitoring, control and shutdown functions. Vendor shall provide hardwired and serial link interface signals for monitoring, control and shutdown functions. Interface with ESD shall be hardwired via Interposing Relay Panel MODBUS interfacing shall be utilized for the serial communication signals	Comply			
3.5	On-line testing facility	-	Required for Generator	Comply			
3.6	Emergency Shut down facility including the shutdown valve and required control valves shall be provided as part of the package	-	Required	Comply			
3.7	Internal Wiring and Terminations	-	Vendor to provide	Comply			
3.8	Panel Painting	-	As per MANUFACTURER's standard	Comply			
3.9	Instrument Specification	-	As per MANUFACTURER's standard	Comply			
3.10	Earthing	-	Vendor to provide a busbar connecting all the conducting material properly earthed with adequate earth cable on the skid baseplate.	Comply			
3.11	Internal Tripping	-	Vendor shall consider internal Tripping signal and control signals within LCP.	Comply			
4.0	COMPLIANCE TO INTERNATIONAL STANDARDS						
4.1	IEC, BS & NFPA, NEMA	-	Required	Comply			
5.0	GENERAL REQUIREMENTS						
5.1	Quality and Vendor data Requirements	-	As per VDRL	Comply			
5.2	Painting	-	As stated above, Before shipment, all equipment's and components of the unit shall be Facility Painted to international standards, marine grade suitable for humid, dusty and saline site conditions,	Comply			
6.0	INSPECTION AND TESTING						
6.1	Insulation Resistance Tests	-	Required	Comply			
6.2	Hydrostatic Testing	-	Required	Comply			
6.3	Load Test	-	Required	Comply			
6.4	Dynamic Load Test	-	Required	Comply			
6.5	FAT	-	Required	Not Applicable			
6.7	On Site Tests (SAT)	-	Required	Comply			
7.0	DOCUMENTATION						
7.1	Documentation	-	As per VDRL	Comply			
7.2	Language	-	English	Comply			
7.3	VENDOR Document Requirement - VDRL	Attachment-09 VDRL This list represents the minimum list of documents. Vendor shall include all required for manufacturing and document which are deemed necessary for quality control and commissioning	Vendor to confirm compliance	Comply			
8.0	SPARE PARTS & SPECIAL TOOLS						
8.1	Pre-commissioning/ Commissioning / Start up spares	-	Required / Vendor to provide, Start-up & Commissioning Spares : As recommended by Vendor with item wise quantity and price break up (To be ordered with the equipment). As a minimum shall include 200% gaskets & 10% bolting (minimum 2 nos. Bolts/4 nuts per size) shall be provided per gasketted internal & external joints (for nozzles with B/F).	Comply as noted. only commissioning spare parts have been offered.			
8.2	Two Years Operation Spares	-	Required - List in SPIR Form to be submitted	Will be offered seperately			
8.3	Special Tools and Tackles for Erection, Pre-commissioning, Commissioning & Operation (if applicable)	-	Vendor to Furnish	will be furnished at later Stage			
8.4	A list of special tools necessary to dismantle the GEG	-	Vendor to Furnish	will be furnished at later Stage			
9.0	OTHER REQUIREMENTS						
9.1	Guarantee / Warranty (Months)	-	12 Months after commissioning	Comply			
9.2	Training Manual & Installation Operation, Maintenance Manuals	-	Required	Comply			
9.3	Supervision for Commissioning at site	-	Per diem rates	Comply			
9.4	Delivery location	-	Construction Site, UAE	Comply			
9.5	Delivery Duration from Purchase Order placed date	weeks	VTA	6-8 Months			
9.6	Fabrication / Inspection & Test Location	-	Vendor to Furnish	Jebel Ali / Hamriya Freezone			
9.7	Local Agent Details	-	Vendor to Furnish	Not Applicable			
9.8	Vendor's Engineering & Procurement Location	-	Vendor to Furnish	Jebel Ali Freezone			
9.9	Preservation	-	Preservation suitable for 6 months storage in marine, hot and humid saline atmosphere	Comply			
9.10	Vendor shall supply Gas Engine Generator (including accessories) in assembled condition. Bidder to confirm.	-	Required	Confirmed			
ABBREVIATIONS : VTA: Vendor to advise TBA:- To be advised VTC: Vendor to comply VTF: Vendor to furnish NA: Not Applicable MOC: Material of Construction							